Requested Features

General

All objects should be selectable.

Tau

Are TauJets the correct collection to use for tau leptons?

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		truth particle
truthParticleLink	vector<ElementLink<DataVector<xAOD::TruthParticle_v1> > >	Link from photon to corresponding truth object
truthType	vector<int>	MCTC (Monte Carlo Truth Classifier) type of matched truth particle
vertexLinks	vector<vector<ElementLink<DataVector<xAOD::Vertex_v1> > > >	Photon -> vertex

AnalysisTauJets[back to top]

Variable Name	Type	Description
charge	vector<float>	Charge of the reconstructed taus (integer, but it can be different from +-1)
EleRNNLoose_v1	vector<char>	Electron discrimination Loose working point
EleRNNMedium_v1	vector<char>	Electron discrimination Medium working point
EleRNNTight_v1	vector<char>	Electron discrimination Tight working point
eta	vector<float>	Reconstructed taus: kinematics
etaFinalCalib	vector<float>	Reconstructed eta from the final tau calibration
etaTauEnergyScale	vector<float>	Reconstructed taus at calo level energy scale: kinematics
isTauFlags	vector<unsigned int>	Set of flags to check if taus has passed some ID/eVeto levels
IsTruthMatched	vector<char>	Reco to truth tau matching

Final States

The following final states should be supported:

- 4 muons (4μ)
- 4 electrons (4e)

BJets (b-tagged jets), LJets (light jets)

Every jet in AnalysisJets has a link to a b-tagging object in BTagging_AntiKt4EMPFlow. That b-tagging object stores three scores from the DL1dv01 neural network tagger:

- DL1dv01_pb — probability the jet came from a **b-quark**
- DL1dv01_pc — probability the jet came from a **c-quark**
- DL1dv01_pu — probability the jet came from a **light quark/gluon**

These three are combined into a single log-likelihood discriminant:

DL1d = ln(pb / (fc * pc + (1-fc) * pu))
Where fc ≈ 0.018 is the fraction of c-jets (a tunable parameter). This gives one number per jet — higher means more b-like.

ATLAS defines standard **working points** (WP) by cutting on this discriminant:

Working Point	Efficiency	DL1d threshold
60% WP	60% b-jet efficiency	~4.93 (tightest)
70% WP	70% b-jet efficiency	~3.56
77% WP	77% b-jet efficiency	~2.51 (recommended)
85% WP	85% b-jet efficiency	~0.92 (loosest)

So at 77% WP: if DL1d > 2.51 → **BJet**, else → **LJet**.

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BTagging_AntiKt4EMPFlow [back to top]

Variable Name	Type	Description
DL1dv00_pb	vector<float>	Posterior score of DL1dv00 tagger (replaced by DL1dv01)
DL1dv00_pc	vector<float>	Posterior score of DL1dv00 tagger (replaced by DL1dv01)
DL1dv00_pu	vector<float>	Posterior score of DL1dv00 tagger (replaced by DL1dv01)
DL1dv01_pb	vector<float>	Posterior score of DL1dv01 tagger
DL1dv01_pc	vector<float>	Posterior score of DL1dv01 tagger
DL1dv01_pu	vector<float>	Posterior score of DL1dv01 tagger

Charge

In a situation where we have a 3-lepton final state, and we're creating a histogram using the invariant mass of one opposite-sign (OS) pair and one same-sign (SS) pair. For example, in an event with $e^+e^-e^-$, we might calculate:

- The invariant mass of the e^+e^- pair (OS).
- The invariant mass of one of the e^-e^- pairs (SS).

These invariant mass values are then used to fill a histogram.

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AnalysisElectrons | AnalysisJets | AnalysisLargeRJets | AnalysisMuons | AnalysisPhotons | AnalysisTauJets | AnalysisTrigMatch | AntiKt10TruthSoftDropBeta100Zcut10Jets | AntiKt4TruthDressedWZJets | BTagging_AntiKt4EMPFlow | CombinedMuonTrackParticles | egammaClusters | EventInfo | ExtrapolatedMuonTrackParticles | GSFConversionVertices | GSFTrackParticles | HardScatterParticles | HardScatterVertices | InDetTrackParticles | Kt4EMPFlowEventShape | MET_Core_AnalysisMET | MET_Truth | MuonSpectrometerTrackParticles | PrimaryVertices | TauTracks | TruthBoson | TruthBosonsWithDecayParticles | TruthBosonsWithDecayVertices | TruthBottom | TruthElectrons | TruthEvents | TruthForwardProtons | TruthMuons | TruthNeutrinos | TruthPhotons | TruthPrimaryVertices | TruthTaus | TruthTop

AnalysisElectrons[back to top]

Variable Name	Type	Description
ambiguityLink	vector<ElementLink<DataVector<xAOD::Egamma_v1> > >	Links Photon<-> Electron when ambiguous
ambiguityType	vector<unsigned char>	Ambiguity (almost surely electron 0 or photon 7/6 (>= rel22/rel21) or ambiguous 1-6/5, (>= rel22/rel21)
author	vector<unsigned short>	Electron, Photon, Ambiguous, Forward Electron
caloClusterLinks	vector<vector<ElementLink<DataVector<xAOD::CaloCluster_v1> > > >	Photon/electron -> Cluster
charge	vector<float>	Electron charge from best track match (+/- 1)

Charge should also be available for muons. Note that charge is not available for jets.

MET(low priority for now)

MET_Core_AnalysisMET provides only the components mpx, mpy, sumet — no pt/eta/phi directly, so MET needs to be reconstructed as pt = sqrt($mpx^2 + mpy^2$), phi = atan2(mpy, mpx),

MET_Core_AnalysisMET[back to top]

Variable Name	Type	Description
mpx	vector<float>	Missing transverse energy core soft term x component
mpy	vector<float>	Missing transverse energy core soft term y component
name	vector<string>	Missing transverse energy core soft term name
source	vector<ULong64_t>	Missing transverse energy core soft term source bitmask
sumet	vector<float>	Missing transverse energy core soft term scalar sum pT

MET_Truth[back to top]

Variable Name	Type	Description
mpx	vector<float>	Truth missing transverse energy x component
mpy	vector<float>	Truth missing transverse energy y component
name	vector<string>	Truth missing transverse energy name
source	vector<ULong64_t>	Truth missing transverse energy source bitmask
sumet	vector<float>	Truth missing transverse energy scalar sum pT

and eta is undefined (MET is transverse-only).

Info from here: https://atlas-physlite-content-opendata.web.cern.ch/